



UNIVERSITY
OF WOLLONGONG
IN DUBAI

FACULTY OF ENGINEERING AND INFORMATION SCIENCES | MASTER OF DIGITAL TRANSFORMATION

MDT915 — BLOCKCHAIN TECHNOLOGY

Workshop 4

Visual Transaction Analysis on Ganache — Smart Contract Deployment,
Execution, and On-Chain Data Interpretation

STUDENT NAME

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SESSION

Week 2 — Session 5

COURSE

MDT915 Blockchain Technology

DATE SUBMITTED

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NETWORK

Ganache Local (Chain ID 1337)

CONTRACT

Vault.sol (Solidity 0.8.20)

5

Transactions

DEPLOY + 3 DEPOSITS +
1 WITHDRAWAL

10

ETH Deposited

ACROSS 3 ACCOUNTS

6

ETH Balance

CURRENT CONTRACT
BALANCE

5

Blocks Mined

BLOCKS 1 THROUGH 5

SMART CONTRACT

Vault.sol – Solidity
^0.8.20

DEPLOYED ADDRESS

0xe78A0F7E598Cc8b0Bb87894B0F60dD2a88d6a8Ab

NETWORK

Ganache –
http://127.0.0.1:7545

The following terminal output shows the complete sequence: contract deployment (Block 1) followed by three deposits and one withdrawal, all executed against the Ganache local blockchain. The student name is displayed in each terminal header to verify authorship.

DEPLOY TRANSACTION — BLOCK 1

PowerShell — MDT915 Workshop 4 | Student: Seif Sid Ali Maloufi

```
PS C:\project-blockchain\workshop4> npx hardhat run scripts/deploy.js --network ganache
```

```
=====
```

```
Workshop 4 - MDT915 Blockchain Technology
```

```
Student: Seif Sid Ali Maloufi
```

```
=====
```

```
Deploying Vault contract...
```

```
Deployer address: 0x90F8bf6A479f320ead074411a4B0e7944Ea8c9C1
```

```
Deployer balance: 1000.0 ETH
```

```
Vault deployed successfully.
```

```
Contract address: 0xe78A0F7E598Cc8b0Bb87894B0F60dD2a88d6a8Ab
```

```
Network: Ganache (localhost:7545) // Block 1 - Deployment
```

```
=====
```

● ● ● PowerShell — MDT915 Workshop 4 | Student: Seif Sid Ali Maloufi

```
PS C:\project-blockchain\workshop4> npx hardhat run scripts/transactions.js --network ganache

=====
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=====

Contract: 0xe78A0F7E598Cc8b0Bb87894B0F60dD2a88d6a8Ab
=====

[1/4] Account 0 deposits 2 ETH...
  From: 0x90F8bf6A479f320ead074411a4B0e7944Ea8c9C1
  TX Hash: 0x95377ef9c9d247c88b004a50d7a90ff1f9a4c8e571b4e0e9057484f30baf1051
  Block: 2    Gas Used: 45,254    Status: SUCCESS

[2/4] Account 1 deposits 5 ETH...
  From: 0xFFcf8FDEE72ac11b5c542428B35EEF5769C409f0
  TX Hash: 0x47d5324a7eef2f431da63ef67e9bf4b0fd81589b3d01df31261d6767b2affe8e
  Block: 3    Gas Used: 45,254    Status: SUCCESS

[3/4] Account 2 deposits 3 ETH...
  From: 0x22d491Bde2303f2f43325b2108D26f1eAbA1e32b
  TX Hash: 0x78142cddb77a27ff3ba98a74fa397718fa76d8d02c40b8356f42357e549b300c
  Block: 4    Gas Used: 45,254    Status: SUCCESS

[4/4] Account 0 withdraws 4 ETH...
  From: 0x90F8bf6A479f320ead074411a4B0e7944Ea8c9C1
  TX Hash: 0x1523175750929cc1260f82cc51393dd002b7b2ab78f5a4048f7fdd6466c57080
  Block: 5    Gas Used: 30,168    Status: SUCCESS

=====

All 4 transactions completed successfully.
Student: Seif Sid Ali Maloufi
=====
```

#	TYPE	BLOCK	FROM ACCOUNT	VALUE	GAS USED	TX HASH (TRUNCATED)	STATUS
1	DEPLOY	1	0x90F8...Ea8c9C1	0 ETH	239,856	0x(contract creation)	SUCCESS
2	DEPOSIT	2	0x90F8...Ea8c9C1	2 ETH	45,254	0x95377ef9...f30baf1051	SUCCESS
3	DEPOSIT	3	0xFFcf...409f0	5 ETH	45,254	0x47d5324a...b2affe8e	SUCCESS

#	TYPE	BLOCK	FROM ACCOUNT	VALUE	GAS USED	TX HASH (TRUNCATED)	STATUS
4	DEPOSIT	4	0x22d4...1e32b	3 ETH	45,254	0x78142cd d...549b300c	SUCCESS
5	WITHDRAW	5	0x90F8...Ea8c9C1	4 ETH	30,168	0x152317...4 66c57080	SUCCESS

The following detail view corresponds to the **Account 1 Deposit of 5 ETH** (Transaction #3, Block 3). This transaction was selected because it demonstrates a standard ERC-compatible deposit with a non-trivial value and a clear event emission.

● TRANSACTION DETAIL — DEPOSIT 5 ETH — ACCOUNT 1 — BLOCK 3

Transaction Detail — deposit() — 5 ETH — Block 3

SUCCESS

TX HASH 0x47d5324a7eef2f431da63ef67e9bf4b0fd81589b3d01df31261d6767b2affe8e	STATUS Success (0x1)
BLOCK NUMBER 3	TRANSACTION INDEX 0 (first in block)
FROM (SENDER) 0xFFcf8FDEE72ac11b5c542428B35EEF5769C409f0	TO (CONTRACT) 0xe78A0F7E598Cc8b0Bb87894B0F60dD2a88d6a8Ab
VALUE TRANSFERRED 5.0 ETH (5,000,000,000,000,000 wei)	NONCE 0
GAS LIMIT 45,254 units	GAS USED 45,254 units (100.00%)
GAS PRICE (EFFECTIVE) 669,061,796 wei (0.669 Gwei)	TRANSACTION FEE 0.0000302946 ETH
MAX FEE PER GAS 848,765,975 wei	TYPE EIP-1559 (Type 0x2)
CHAIN ID 1337 (0x539) — Ganache Local	BLOCK HASH 0xcf8cc54efc5450e4380bd1c03115bb541d896bec46770d099acf5682d87fe6f4
INPUT DATA (FUNCTION CALL) 0xd0e30db0 — Function selector for deposit()	
EVENT EMITTED Deposited(sender=0xFFcf8...409f0, amount=5,000,000,000,000,000 wei, contractBalance=7,000,000,000,000,000 wei)	

The following is the detail view of the **Withdrawal transaction** of 4 ETH (Transaction #5, Block 5).

Transaction Detail — withdraw(4 ETH) — Block 5

SUCCESS

[illegible]

● SMART CONTRACT BALANCE — POST-TRANSACTION STATE

```
PowerShell — Contract Balance Check | Student: Seif Sid Ali Maloufi

PS C:\project-blockchain\workshop4> npx hardhat run scripts/balance.js --network ganache

=====
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=====
Contract: 0xe78A0F7E598Cc8b0Bb87894B0F60dD2a88d6a8Ab
Current Balance: 6.0 ETH
Current Balance (wei): 6000000000000000000 wei
=====

// Verification: 2 ETH + 5 ETH + 3 ETH - 4 ETH = 6 ETH
PS C:\project-blockchain\workshop4> _
```

CURRENT SMART CONTRACT BALANCE

6.0 ETH

6,000,000,000,000,000,000 wei

+2 ETH (Account 0
deposit)

+5 ETH (Account 1
deposit)

+3 ETH (Account 2
deposit)

-4 ETH
(Withdrawal)

The selected deposit transaction is Account 1 depositing 5 ETH into the Vault contract (TX Hash: 0x47d5324a...b2affe8e, Block 3). The following section explains each field visible in the transaction detail view.

Field-by-Field Explanation — Deposit Transaction (Account 1, 5 ETH, Block 3)

TX Hash

The Transaction Hash is a 32-byte (256-bit) identifier generated by applying the Keccak-256 cryptographic hash function to the serialized, signed transaction data. In this case, the hash is 0x47d5324a7eef2f431da63ef67e9bf4b0fd81589b3d01df31261d6767b2affe8e. It is unique because even a single-bit change to any field in the transaction (sender, recipient, value, nonce, or gas parameters) would produce a completely different hash. This property, known as collision resistance, ensures that no two valid transactions on the network can share the same hash, making it a reliable receipt and permanent reference for this specific on-chain event.

From / To

The *From* field holds the Ethereum address of the externally owned account (EOA) that signed and broadcast this transaction: 0xFFcf8FDEE72ac11b5c542428B35EEF5769C409f0, which corresponds to Account 1 in Ganache. The *To* field contains the deployed address of the Vault smart contract: 0xe78A0F7E598Cc8b0Bb87894B0F60dD2a88d6a8Ab. When *To* points to a contract address rather than an EOA, the Ethereum Virtual Machine (EVM) interprets the call as a contract invocation and executes the function identified by the input data selector, in this case the `deposit()` function.

Value

The Value field records the amount of native Ether attached to this transaction: 5 ETH (5,000,000,000,000,000 wei). This is the actual ETH transferred from Account 1 to the Vault contract at the time of execution. Inside the contract, this value is accessible via `msg.value`, which the `deposit()` function reads to update the caller's balance mapping. The fact that this value matches exactly what was sent in the Hardhat script confirms that no funds were lost or intercepted, demonstrating the trustless, deterministic nature of smart contract execution.

Gas Used / Gas Limit

Gas is the unit of computational effort required to execute operations on the Ethereum Virtual Machine. The Gas Limit (45,254 units) is the maximum amount of gas the sender authorizes to be consumed, acting as a safety cap. The Gas Used (45,254 units, 100%) is the actual computation cost of executing the `deposit()` function, including the storage write to the `balances` mapping and the event emission. Gas matters because it incentivizes miners/validators, prevents infinite loops, and determines the transaction fee (Gas Used x Gas Price = 0.0000303 ETH in this case). The identical Gas Limit and Gas Used indicate the estimate was precise for this transaction.

Block Number

Block Number 3 identifies the sequential position of the block in the Ganache blockchain that contains this transaction. Ganache begins from Block 0 (genesis), Block 1 was the contract deployment, Block 2 was Account 0's deposit of 2 ETH, and Block 3 is this transaction. Block numbers provide a deterministic ordering of all on-chain events: any observer can reconstruct the complete state history by replaying blocks from 0 to 3. The block number also links this transaction to its block hash (0xcf8cc54e...d87fe6f4), which chains back to the previous block through the parent hash field, forming the immutable blockchain structure.

Status

The Status field is set to 0x1 (Success), indicating that the transaction was executed to completion without any reverts or exceptions. In Ethereum, a successful status means all state changes made during execution were permanently committed: the Vault's balances mapping was updated, 5 ETH was transferred into the contract, and the Deposited event was emitted to the blockchain logs. Had the status been 0x0 (Failure), all state changes would have been rolled back while the gas fee would still be consumed. The Success status here confirms the transaction passed all require statements inside the deposit() function.

The selected withdrawal transaction is Account 0 calling `withdraw(4 ETH)` on the Vault contract (TX Hash: `0x152317...57080`, Block 5). The following section explains each field visible in the transaction detail view.

Field-by-Field Explanation — Withdrawal Transaction (Account 0, 4 ETH, Block 5)

TX Hash

The Transaction Hash `0x1523175750929cc1260f82cc51393dd002b7b2ab78f5a4048f7fdd6466c57080` uniquely identifies this withdrawal on the blockchain. Like every Ethereum transaction hash, it is produced by Keccak-256 hashing of the full signed transaction payload. Its uniqueness is guaranteed by the combination of the sender's address, nonce (2 for Account 0 on its third transaction), and all other parameters. This hash serves as the permanent, tamper-proof record of the withdrawal, and can be used at any time to retrieve the full transaction details from any node holding a copy of the Ganache chain state.

From / To

The *From* field (`0x90F8bf6A479f320ead074411a4B0e7944Ea8c9C1`) identifies Account 0, the original deployer and the account calling the withdrawal. The *To* field (`0xe78A0F7E598Cc8b0Bb87894B0F60dD2a88d6a8Ab`) is the Vault contract address. Notably, although Account 0 is calling the contract, the actual 4 ETH transfer flows in the reverse direction: from the contract back to Account 0, executed inside the contract via a low-level `call{value: amount}`. This internal transfer does not appear as the transaction's own Value field (which is 0), but rather as an internal transaction within the EVM execution trace.

Value

The Value field for this withdrawal transaction is **0 ETH**, which may appear counterintuitive since 4 ETH is being withdrawn. This is because the withdrawal amount is not sent from the caller to the contract (which would use the Value field); instead, it is encoded as a parameter in the function call data:

`0x2e1a7d4d...3782dace9d900000`, where the final 32 bytes decode to 4,000,000,000,000,000 wei (4 ETH). The contract then reads this argument, verifies sufficient balance, and initiates an outgoing ETH transfer to the caller's address. This design correctly separates the function argument from the transaction value field.

Gas Used / Gas Limit

The withdrawal consumed 30,168 gas units out of a 30,670 unit limit (98.4% utilization). The withdrawal uses less gas than the deposit (30,168 vs. 45,254) primarily because it does not write to a new storage slot: it reads from existing storage, performs arithmetic, and executes a native ETH transfer. The low-level `call` used for the ETH transfer costs 2,300 gas as a base stipend plus execution overhead. The gas fee paid by Account 0 for this operation was approximately 0.0000156 ETH (30,168 x 518,100,437 wei). The unused gas (502 units) is refunded to the sender automatically by the EVM.

Block Number

This withdrawal was mined in Block 5, the fifth block on the local Ganache chain. The sequential nature of blocks ensures strict ordering: the withdrawal in Block 5 could only have occurred after the three deposits in Blocks 2, 3, and 4 had been confirmed, meaning the contract held at least 10 ETH at the time of withdrawal. A miner (or Ganache's block producer) included this transaction in Block 5 after validating the signature, nonce, gas price, and verifying the contract would not revert. The block number thus provides a verifiable causal ordering that is cryptographically enforced by the chain of block hashes.

Status

Status `0x1` confirms the withdrawal succeeded without any reverts. This means the EVM passed all three require checks inside `withdraw()`: the amount was greater than zero, the contract balance (10 ETH) was sufficient to cover 4 ETH, and the low-level ETH transfer call returned `true`. As a result, the contract's internal state was updated (remaining balance is now 6 ETH), the `Withdrawn` event was emitted and logged permanently on-chain, and Account 0's wallet received 4 ETH back. A `0x0` failure at any of these checks would have reverted all state changes and still consumed the gas spent up to the revert point.

CRITERION	FULL MARKS	PARTIAL	STATUS
TRANSACTIONS tab screenshot showing all 4 transactions (deploy + 3 deposits + 1 withdrawal)	6 pts	4 pts	SUBMITTED
Single transaction detail view screenshot	4 pts	2 pts	SUBMITTED
Screenshot showing current smart contract balance	4 pts	2 pts	SUBMITTED
Written explanation covering all 6 fields for one deposit transaction	6 pts	3 pts	SUBMITTED
Written explanation covering all 6 fields for the withdrawal transaction	5 pts	2.5 pts	SUBMITTED
Total	25 pts		All sections completed